

AMD DEVELOPER HACKATHON · TRACK 3 (UNICORN)

Caridence

Walk around the car. Get the truth.

The **problem**

Every used-car sale, rental return, and insurance claim starts the same way: **someone walks around the car and writes down what they see.**

- Slow and inconsistent — depends who's looking
- Misses damage, especially subtle scratches and cracks
- No evidence trail when a dispute arises

WHAT IT DOES

A phone walkaround → a **cited** damage report

Point a phone at the car, walk around once. Caridence returns every finding **cited to the exact frame and box**, graded for severity, with an estimated repair cost — and blurs license plates automatically.

How it **works**



Detector guarantees we don't miss damage · verifier keeps false alarms out.

Fine-tuning is the **whole story**

0.00

Base model, zero-shot

Can't produce grounded damage JSON

0.56

Fine-tuned F1 (held-out CarDD)

Small open model → viable product

96%

Presence recall

Detector, 374-image test set

Small, specialized, self-hosted open models — no frontier-API dependency.

Runs on **AMD ROCm**

66

FPS · single pass

Radeon gfx1100, ROCm 7.2, torch 2.9.1

26

FPS · high-recall TTA

Faster than the video was filmed

1x

Same path → MI300X

No source changes across AMD parts

Benchmarked live on AMD hardware · ROCm training + serving · Gemma-via-Fireworks option.

Open & reproducible

- **Public MIT repo** — github.com/dholzric/caridence
- **Containerized** — docker build && docker run, no GPU needed
- **Weights + benchmark in-repo** — reproduce the AMD numbers
- **Privacy by default** — plates auto-redacted before storage

Who needs **this**

- **Rental fleets** — millions of returns a year, inspected in seconds
- **Dealers & marketplaces** — condition reports buyers can trust
- **Insurers** — claims triaged in minutes, with an evidence trail

One 60-second phone walk → a report all of them can rely on.

BUILT ON OPEN MODELS & AMD

Caridence

caridence.com · github.com/dholzric/caridence